Q-INTEL

QUICK-COMMERCE DELIVERY INTELLIGENCE

BUSINESS SOLUTION

A practical operating model for predicting delivery delays, understanding demand zones and improving delivery operations.

THE BUSINESS PROBLEM

Unpredictable delivery times, late-order cancellations, uneven rider workload and uncertainty about where to prioritise dark-store capacity.

THE Q-INTEL SOLUTION

A decision-support workflow combining ETA prediction, late-risk alerts, geographic demand analysis, courier segmentation and prioritised operational actions.

1. Solution objective

Q-INTEL is designed to help an operations team answer four connected questions:

- **How long** may an order take?
- **Which orders** are at risk of breaching the service-level agreement (SLA)?
- **Where** are demand concentrations and candidate service zones?
- **Which courier groups** may need operational support?

The solution converts historical order data into understandable signals and recommended actions. It is intended to support human decisions, not automatically dispatch riders, change customer promises or approve new store locations.

2. Business outcomes targeted

Outcome	How the solution supports it	Measure for a pilot
Earlier risk awareness	Surface orders with higher predicted late probability for review.	Late rate; late-order recall; alert precision
More realistic customer ETA	Provide a predicted delivery duration and, if validated, an uncertainty range.	MAE in minutes; interval coverage
Evidence-led zone planning	Compare demand, service time and late rate across geographic clusters.	Orders/zone; zone late rate; cost/order
Better courier support	Use rider segments to inform coaching and workload review.	Segment late rate; workload balance

Scope and data note

The challenge dataset is a public food-delivery dataset, not a native grocery quick-commerce dataset. Restaurant coordinates are treated as store-location proxies. It does not contain all required grocery inventory, picking, dark-store rent, road-network, capacity or actual dispatch data. Therefore, Q-INTEL provides a prototype solution and candidate insights that require validation with real quick-commerce data.

3. Proposed End-to-End Business Workflow

The solution follows a controlled flow from source data to operational decisions. Each stage has a defined input, output and validation responsibility.

Stage	What happens	Output / control
1. Data intake	Load the approved historical delivery file and verify schema, row count and target field.	Versioned raw input; schema check
2. Data quality	Profile missingness, duplicates, invalid GPS, malformed times, category inconsistencies and suspicious values.	Quality report; documented retain/flag/exclude decisions
3. Feature engineering	Parse delivery duration; derive order time/calendar fields; calculate Haversine distance for valid coordinate pairs.	Model-ready features; GPS-valid flag
4. Model training	Use chronological or otherwise justified holdout splitting; fit preprocessing only on training data; compare baselines and candidate models.	Versioned model bundles and evaluation metrics
5. Operational scoring	For a new eligible order, estimate ETA and late probability; apply the validation-selected threshold.	ETA, probability, risk category and explanation
6. Zone and courier analysis	Cluster valid geographic demand points and aggregate courier performance features where available.	Zone profiles; courier segments
7. Decision layer	Combine model signals and observed business patterns into a ranked review list and three recommendations.	Manager-facing actions and KPIs
8. Demonstration	Expose an order-input workflow and dashboard in Streamlit; display assumptions and model version.	Reproducible demo; human review

Decision workflow for a new order

Input	Model / rule	Output for the operator
Order details and valid prediction-time features	ETA regression model	Predicted delivery time in minutes
Same eligible feature set and SLA label definition	Late-risk classifier + validation-selected threshold	Late probability, risk flag and review priority
Location and zone assignment, if supported	Geographic zone lookup / clustering output	Relevant zone demand and service context

Human-in-the-loop operating principle
A high-risk flag should trigger a review, not an automatic rider reassignment. The operator checks store workload, distance, current courier availability and customer-service options. These real-time operational inputs are not fully represented in the historical dataset and must be connected before live dispatch decisions can be automated.

Prediction timing and leakage control
Define exactly when the prediction is made. Do not use pickup or delivery-completion information for a prediction made at order placement. Rider and dispatch attributes may be used only in a later-stage model if they are available at that decision time. Weather and traffic fields also require a clear availability-time check.

4. Operational Playbooks

The following playbooks show how an operations manager can use Q-INTEL outputs. They are proposed workflows; they do not imply that interventions have already been deployed or that savings have been measured.

Playbook A — Late-order prevention

Step	Action	Responsible role / evidence
1. Detect	Review orders whose predicted late probability crosses the threshold selected on validation data.	Operations analyst; classifier output
2. Diagnose	Check distance, demand concentration, weather/traffic signal availability and store workload.	Shift manager; current operational data
3. Decide	Choose an approved response: monitor, contact customer, review workload or request available support.	Operations manager; documented policy
4. Measure	Compare late rate, recall, precision, false-alert volume and intervention cost against a baseline.	Analytics team; controlled pilot

Playbook B — Dark-store / demand-zone planning

Step	Action	Decision safeguard
1. Map	Display all discovered geographic clusters, demand points and cluster centres on an interactive map.	Use valid coordinates; do not show invalid GPS as real locations.
2. Compare	Rank zones by demand volume, average delivery duration and late rate against network benchmarks.	Use actual cluster summary; show sample size.
3. Investigate	Check candidate locations against rent, inventory, staffing, road access, capacity and service radius.	These factors require additional business data.
4. Pilot	Run a small capacity or micro-fulfilment pilot before a permanent site decision.	Measure cost per order and service-level improvement.

Playbook C — Courier performance support

Step	Action	Decision safeguard
1. Segment	Group couriers using defensible rider-level aggregates such as experience, rating, average duration, late rate and deliveries/day.	Use only available and appropriate rider records.
2. Profile	Describe each group using actual cluster averages, counts and uncertainty.	Do not assign labels before inspecting segment statistics.
3. Support	Consider coaching, workload balancing or roster review for groups needing support.	Avoid punitive decisions from unadjusted observational data.
4. Review	Track segment metrics over time and check whether support is associated with improvement.	Control for route, demand and shift differences.

Operational dashboard design
The Streamlit demonstration should present: an executive overview; new-order ETA and late-risk prediction; zone intelligence with a map and zone comparison; courier segment profiles where available; and a model/explanation panel. Every chart should identify its data period, SLA definition, model version and whether the figure is observed, predicted or illustrative.

5. Business Rules, Governance & Implementation Plan

A. Decision rules

Rule	Recommended policy
SLA definition	Use the challenge-approved SLA. The current project direction has referred to a 30-minute late label; confirm this against the final executed notebook and organiser definition before release.
Late-risk threshold	Select on validation data using the declared business cost ratio; freeze before final test evaluation. Do not tune on the test set.
Risk categories	Define labels from the saved threshold and documented escalation rules. Display probability as a model estimate, not certainty.
Zone priority	Use a transparent score or ranked evidence table based on demand, service performance and data quality. Do not equate a cluster centre with a recommended store site.
Courier segment names	Assign human-readable names only after reviewing segment sizes and actual performance profiles.
Recommendations	Every recommendation must link an observed finding → supporting evidence → proposed action → pilot KPI.

B. Governance and risk controls

Risk	Control
Data leakage	Use prediction-time feature rules; exclude post-outcome fields; use train-only preprocessing.
Biased / incomplete geography	Flag invalid GPS; report excluded coordinate counts; avoid presenting approximate clusters as service boundaries.
Class imbalance	Report precision, recall, F1, ROC-AUC and PR-AUC where appropriate; do not rely on accuracy alone.
Uncalibrated probabilities	Validate calibration before treating risk percentages as operational probabilities.
False alerts	Track alert volume and precision; select threshold using business costs and pilot feedback.
Causal overclaim	Describe EDA and feature importance as associations, not proof that a factor causes delay.
Privacy / fairness	Limit access to rider-level information; review fairness and avoid punitive use without contextual evidence.

C. Phased implementation plan

Phase	Deliverable	Exit criterion
1. Validate	Freeze notebook, data definitions, feature list and selected models.	Reproducible run; documented metrics
2. Demonstrate	Streamlit order scoring, zone map and model comparison.	Inputs validated; outputs traceable to saved models
3. Shadow pilot	Score historical or live-like orders without changing dispatch.	Monitor drift, calibration, false alerts and data availability
4. Controlled pilot	Test an approved review workflow in selected shifts/zones.	Compare treatment vs control using pre-agreed KPIs
5. Scale decision	Evaluate service and cost impact; add production integrations only if justified.	Business owner approval and monitoring plan

6. Three Prioritised Business Actions

The challenge asks for exactly three prioritised management actions. These recommendations are deliberately framed as pilots, because the historical proxy dataset cannot establish actual operational savings or prove that a new store or staffing change will work.

Priority	Recommendation	Evidence to attach	Success measures
1. Protect service levels	Introduce a human-reviewed late-risk queue for orders flagged by the validation-selected classifier. Review workload, distance and available support before deciding on an intervention.	Test-set precision, recall, F1/PR-AUC; threshold and cost assumption; pilot baseline.	Late-order rate; late recall; alert precision; false-alert workload; intervention cost.
2. Investigate high-priority zones	Use geographic cluster demand and service metrics to select candidate areas for a feasibility study. Validate rent, capacity, inventory, road access and customer density before any site decision.	All-zone map; cluster count and validation; demand volume; average duration; late rate; GPS quality.	Orders served; delivery radius; on-time rate; cost per order; capacity utilisation; site economics.
3. Improve courier support and shift planning	Use courier segments and time-of-demand patterns to identify potential coaching, workload balancing or peak-period roster pilots.	Rider segment sizes and profiles; shift/time analysis; comparable route and demand conditions.	Late rate by segment; workload balance; rating trend; rider retention/safety indicators.

Pilot measurement framework
Before any intervention, define a baseline period, treatment group, comparison group, eligibility rules and minimum sample size. Compare changes in late rate, ETA error, cancellations (if available), cost per order, courier workload and customer outcomes. Where possible, use a test/control design and account for differences in demand, distance, weather and shift.

Limitations that management must understand

- The supplied data represents food delivery and uses restaurants as store proxies; it does not fully represent grocery picking, inventory availability, dark-store economics or quick-commerce fulfilment.
- Historical relationships are observational and do not establish causation.
- Geographic clusters are analytical groupings, not verified delivery territories, road routes or optimal store locations.
- The 5:1 missed-late versus unnecessary-intervention cost ratio is an assumption until replaced with measured company costs.
- An ETA uncertainty band must be checked for actual coverage and width on held-out data before being displayed as reliable.
- Live deployment requires current traffic, courier availability, store workload, model monitoring, privacy controls and integration testing.

Conclusion

Q-INTEL provides a practical decision-support blueprint that connects delivery ETA estimation, late-order risk, geographic demand intelligence and courier segmentation. Its business value should be established through a controlled pilot with genuine quick-commerce data and measured outcomes. The immediate deliverable is a reproducible prototype that helps managers ask better questions, prioritise investigation and evaluate operational interventions with evidence.

Submission handover checklist
Attach the final runnable notebook/source code, complete model comparison table, six-page business report, this five-page business solution, working Streamlit demonstration and README. Confirm that the selected models, SLA, thresholds, cluster outputs and metric values are consistent across all files.